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Chapter 01 - Problem Statement

In Sri Lanka, Dengue fever appears to be one of the biggest health problems affecting the public every year. The Ministry of Health's Epidemiology Division estimates that in 2023, there were lots of dengue cases in major cities like Colombo, Gampaha and Kandy. Citizens generally have limited ability to access up-to-date, relevant dengue risk information, despite various awareness efforts. Typical awareness-raising approaches, including reports from the government, public messages, and advertisements, are often reactive and impersonal. As an outcome, people often know about epidemics once the diseases are already in. Due to this, precautions are usually taken after epidemics have started to expand.

The willingness of both citizens and the government to engage in early preventive steps is restricted by the lack of forecasting, data-driven approaches. Epidemics of dengue are getting more prevalent and unpredictable as a result of poor disposal of waste, rising urbanisation, and changing climates, such as heavy rainfall. Mosquito lifecycle patterns are greatly impacted by climate change, especially variations in humidity and temperature. Many locals are uninformed of the manner in which the immediate surroundings support mosquito breeding, resulting in repeated outbreaks in the same areas. For early identification, current techniques, however, exclude data analysis, weather info and community reports.
The value of reliable and easily available health data was further emphasised during the COVID–19 pandemic.

By creating a smart, user-friendly mobile app that employs AI and machine learning models to determine dengue risk levels in real time, my project tries to address these problems. In order to motivate prevention tips, among users, it will also contain public reporting dashboards, localized alerts, interactive mapping, prevention guidance, and awareness information.

With the goal of decreasing rates of infections, increasing public awareness, and improving national health resilience, this approach eventually aims to change Sri Lanka's dengue prevention efforts to respond to active, data-informed methods.

Chapter 02 – Project Description

2.1 Overview:

This task's primary objective aims to develop a mobile app that uses AI models to estimate dengue risk levels across Sri Lankan districts. The app will warn users about the danger levels they are in and include a prevention section with wellness tips.
By using data-driven solutions, it tries to improve healthcare.

2.2 Background and Rationale:

In tropical countries like Sri Lanka, the dengue virus, a frequent mosquito–carried disorder, is extensive, particularly in the monsoon season. Even though they lack in forecasting safety systems, health organizations give regular updates. Data analytics and AI advancements might contribute to the forecasting of viruses. Sri Lankan efforts to manage dengue may be improved through mobile apps and automated digital tolls.

2.3 Project Objective:

The purpose of the project plans to create a machine learning algorithm which employs health and weather data to forecast dengue risk in Sri Lanka. Furthermore, it will also develop a user reporting system, a mobile-friendly dashboard, public awareness initiatives and prevention strategies for health departments.

2.4 Scope:

The main objectives of this project are the progress, execution and analysis of a mobile software concept that predicts and displays dengue risk in particular Sri Lankan areas.
System Architecture: 1. Backend layer 2. AI layer 3. UI layer

Project Inclusions:

- Development of Dengue Risk Prediction model with AI
- Integration for managing data with a cloud-hosted database. (MongoDB, Supabase, PostgreSQL)
- Implementing dynamic dashboards and interactive map visualizations.
- User reporting and notification system for the community.
- Testing and setting up on both iOS and Android systems.

Exclusions:

- At this level, there is no direct link with medical or hospital data systems.
- No private medical data is collected, expected for user-generated reports or basic locations.
- The project relies primarily on secondary and open-sourced data instead of clinical testing.

2.5 Expected Outcomes:

- A simple, straightforward mobile application for forecasting dengue risk.
- A model based on AI that could provide real-time risk warnings.
- Improved public engagement and understanding in preventing dengue through reporting and alerts.

Project Key Terms*: Dengue risk prediction, Machine Learning, Mobile application, Data visualization, Community health awareness*

Chapter 03 - Research Gap

Studies on dengue prediction using AI remain academic concepts which cannot be mobile accessible to the public. Sri Lankan government dashboards are online currently without dynamic forecasts.

For predicting dengue epidemics in Sri Lanka, current models, including the ones designed by (Uduwanage, 2025) , apply machine learning methods such as SARIMAX to predict dengue incidences over 09 districts in Sri Lanka, analyzing humidity, temperature and precipitation.. These simulations, nevertheless, are mainly academic types and lack connectivity with public mobile applications.

Similarly, (Nasir, 2024)applied decision trees and neural networks to build an AI–based dengue forecasting model in Indonesia. Although the model had a high amount of predictive capacity, it was simply an academic model which wasn't yet linked into a community-wide app or real-time healthcare system.

High prediction precision was discovered in this contrasting study of different ML algorithms. It, however, did reach its conclusion that usage and implementation models integrating with health systems were rarely achieved. (Zhao, 2020)

The research suggests a methodological gap in successful algorithms because they usually overlook environmental and social variables and specific data, which decreases their usability and accuracy.

Frequently, current forecasting techniques cannot be implemented for ongoing user-friendly execution. As an example, (Seema Patil, 2021) built a climate-based dengue early detection model for India, and they discovered that the reliability of the model was limited by the lack of regional sociodemographic data and contradictory climatic records.

To improve dengue detection in Sri Lanka, (Mo-Buzz, 2019) merged citizen reporting with mobile technology. Its lack of automatic risk signals and statistical analysis restricted its capacity for active protection, but it was effective in involving the general public. Although the health departments' dashboards contain valuable instances in addition, these are static web-based websites lacking active or predictive functions.

These cases show a technological and knowledge gap: there are many algorithms available, though they have not yet been implemented for ongoing, customer–facing deployment.

Global models developed for tropical environments cannot be instantly used in Sri Lanka without localized validation. Research by (Seema Patil, 2021) for example, used models and data particular to those countries, which could not take into account Sri Lanka's specific weather and social–environmental features.

This underlines a contextual gap: Here no regional models which consider regional factors like density, weather patterns and healthcare systems.

Chapter 04 – Requirement Analysis

Identifying the desired behavior, restrictions, and efficiency criteria recommended that the Dengue Risk Awareness and Prediction Mobile App requires an extensive requirement analysis process.

4.1 Requirement Gathering Methods:

The requirements of the project are going to be collected and confirmed through a range of techniques:

Interviews and Discussions: Interviews and discussions were conducted with the students, public, and community health workers, to find out further about how dengue info currently appears and is easily available.

Observation: Studies on how people nowadays follow dengue epidemics via official sites or media reports.

Online surveys: Provided via online platforms to obtain information about bugs in usability, desired attributes, and alert settings.

Review of competing systems: Analyzing equivalent area disease monitoring procedures to discover gaps and prospective modifications.

Analyze existing documentation: To find any gaps, check Ministry of Health announcements, dashboards, and standards set by WHO.

4.2 Functional Requirements:

Table 1: Shows the functional requirements

Requirement ID	Description
01	A live map showing dengue risk classes at an area is going to be supplied by the system.
02	According to the user's current location, the system will figure out the dengue risk levels.
03	Once someone enters a high–risk location, the system will promptly send warning alerts.
04	Admins will be enabled to review and evaluate aggregated reports from the user via the system.
05	Users will be enabled to submit reports of suspected dengue conditions or areas of mosquito breeding via the system.

06	For the purpose of showing patterns, the system will maintain, upgrade previous dengue incident data.
07	The system will present health suggestions, signs, and protection tips.

4.3 Non-Functional Requirements:

Table 2 : Shows the non - functional requirements

Requirement ID	Description
01	For maximum accessibility, the system must respond to user interaction in 3-5 seconds.
02	Users of all ages will find the interface simple to operate, simple to solve and natural.
03	To secure user information, the system need to include safe authentication methods.
04	To guarantee secure accessibility, the system must have an average uptime of 95%.
05	Throughout dengue peak seasons, the system must be adaptable to handle greater usage.
06	Encrypting data must be applied by the system to secure the user's location.
07	The app needs to function properly on both iOS and Android systems.

4.4 User Requirements:

Table 3: Shows the User requirements

User type	Requirements
General user	Should be capable of easily and quickly viewing dengue risk data.
General user	Once danger levels increase, automatic proactive messages need to be delivered to the users.
Public members	Reports on areas of breeding ought to be provided to the people.
Health authorities	Tracking aggregated data trends from reports must be available.

4.5 Software Requirements:

The system assets will be required to build the system effectively.

Frontend Development:

Flutter or React Native (For universal compatibility)

Backend Development:
Flask API Routes or Next.js (For ML integrations and data analysis)

Database:
MongoDB, Supabase, PostgreSQL (For secure data kept in the cloud)

Mapping and Visualization:
Mapbox API (for danger heatmaps) or leaflet.js

Machine learning tools:
Python, pandas, scikit-learn and TensorFlow (For time series forecasting)

Notification Services:
Firebase Cloud Messaging (For warnings and awareness tips)

4.6 Hardware Requirements:

Table 4: Shows the Hardware requirements

Hardware Component	Specification
Laptop	8GB Memory, 256GB Flash Drive, i5 processor or higher
Mobile testing device	iOS or Android (for UI verification or testing)
Network access	Continuous access to the internet for deployment, testing and API data

4.7 Knowledge Requirements:

- Machine learning model training and development.
- Security practices and Cloud deployment.
- API integration.
- UI/UX design concepts.
- Data visualization.

Chapter 05 – Finance

Table 5: Shows the financial plan of the project

Item	**Description**	**Estimated Cost (LKR)**	**Cost Type**
Development cost	For development and testing	10,000	Direct
Internet and hosting	Supabase backend servers and cloud storage	3000	Indirect
Map API	MAPbox API	2000	Direct
Testing virtual	Telephones and an emulator	5000	Direct
Other	Presentation, printing and documentation	2000	Indirect
Total Estimated Cost:		**22,000**	

Justification:

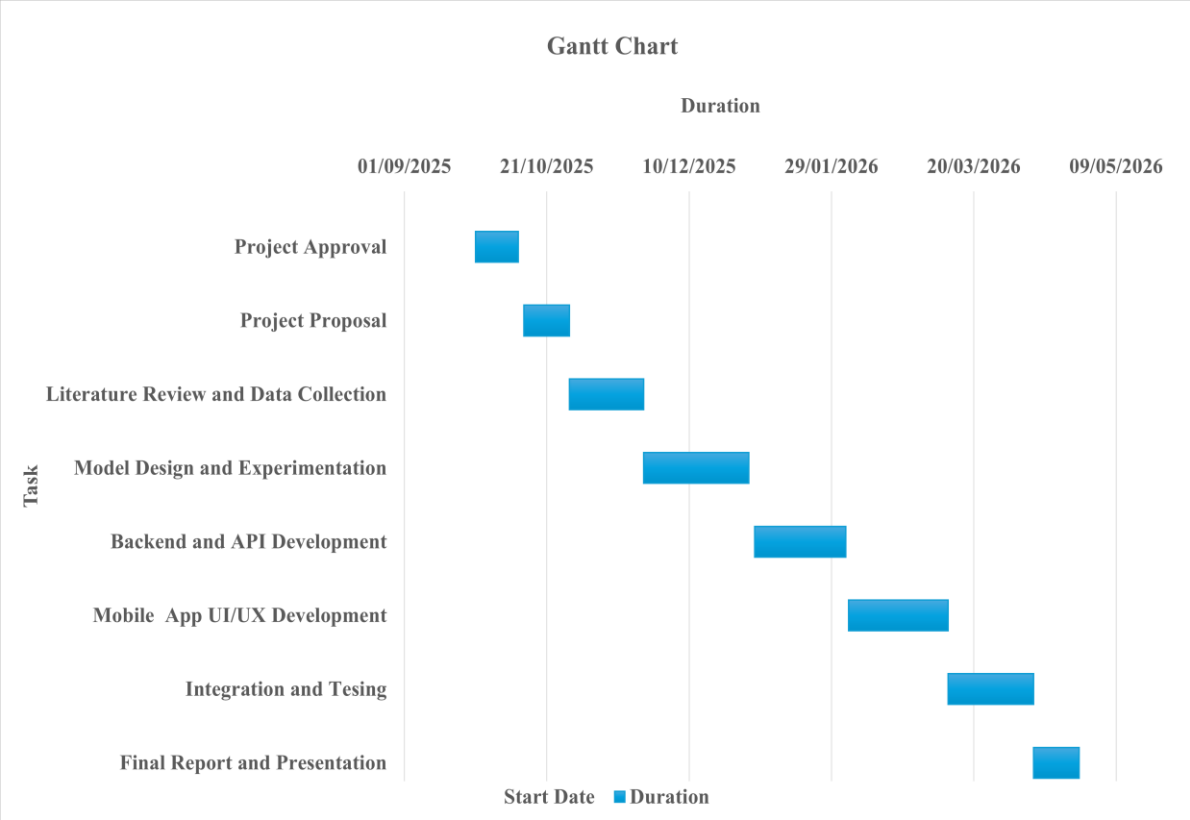
The planned 22,00LKR contains costs for testing tools, cloud hosting, developer labor and APIs. These are needed to make sure that the app functions effectively throughout a range of systems, offers accurate dengue risk visualization, and delivers a professional user experience. Reporting on projects and continuity is provided by additional costs like the internet and paperwork.

Chapter 06 – External Organizations

Table 6: Shows the external organizations

Role	Name / Organization	Description
Supervisor	Mr. Gayan Perera	Ensures conformance with academic norms and gives leadership, methodology, and monitoring support for all activities.
External advisor	Mr. Malith Senanayake Project Manager SRIQ Corporation	Gives technical advice on codes, implementation methods, and AI modelling.
Community partner	Local health clinics	Helps in user testing, gives comments on accessible features and confirms the functionality of apps.

Chapter 07 – Time Frame / Timeline



[Source: Author]

Figure 1: This Gantt chart shows the time frame of the project

Task	Start Date	End Date	Duration
Project Approval	26-09-2025	10-10-2025	15
Project Proposal	13-10-2025	28-10-2025	16
Literature Review and Data Collection	29-10-2025	23-11-2025	26
Model Design and Experimentation	24-11-2025	30-12-2025	37
Backend and API Development	02-01-2026	02-02-2026	32
Mobile App UI/UX Development	04-02-2026	10-03-2026	35
Integration and Tesing	11-03-2026	09-04-2026	30
Final Report and Presentation	10-04-2026	25-04-2026	16

Figure 2: This is the Excel table of the time frame

[Source: Author]

The Gantt chart excludes weekends and university holidays to reflect a realistic working schedule.

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